



useGalaxy.eu

Data-Driven Research for Everyone with the European Galaxy Instance!

Galaxy is an **open, web-based** platform for data driven research.

Accessible

Users can easily configure and run tools without the need to write code, all via a user-friendly web-based interface.



Reproducible

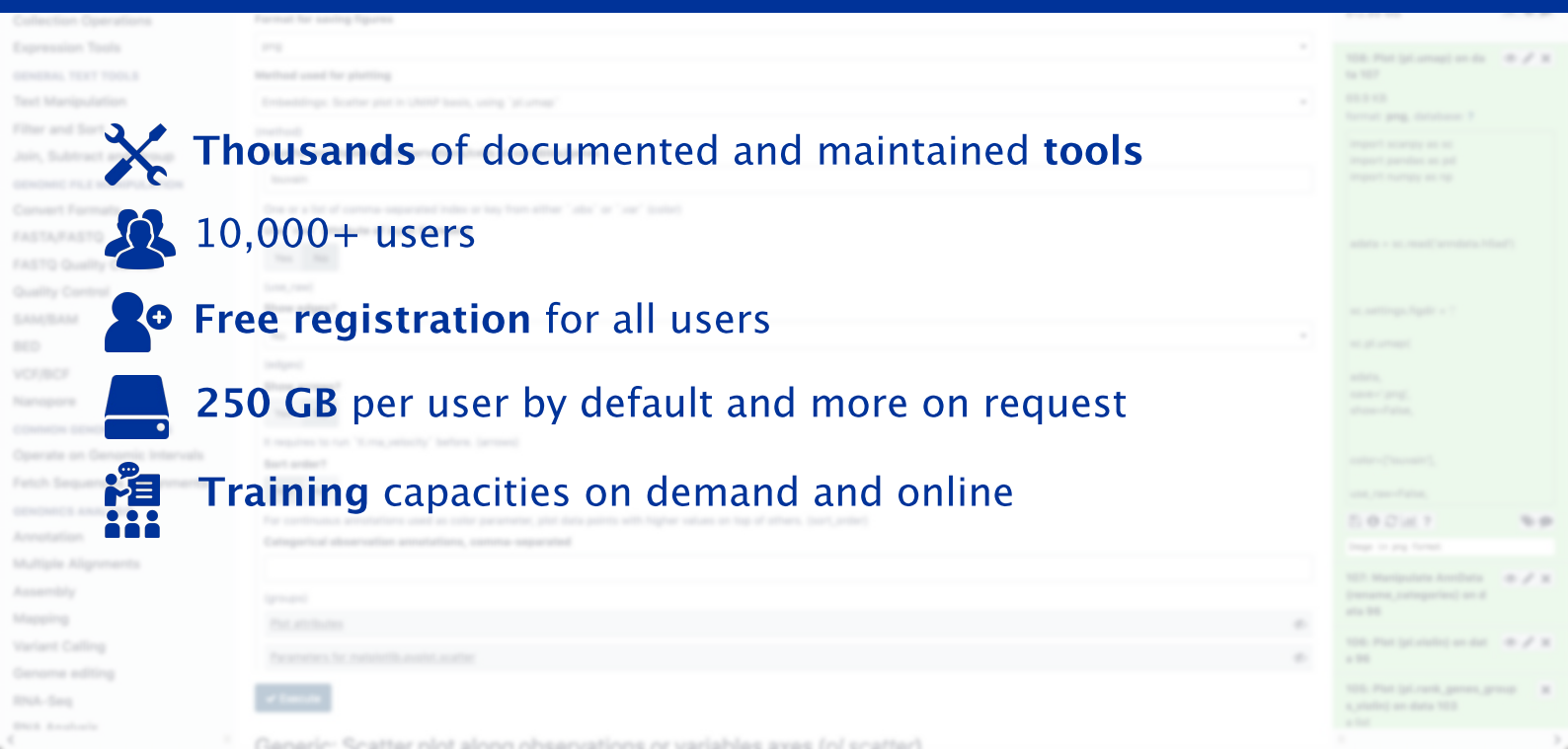
Galaxy captures the full metadata required so that any user cannot only understand but also reproduce the complete computational analysis.

Transparent

Users can extract, share and publish analyses via interactive, web-based documents to add user-supplied annotation.

Scalable

Galaxy scales for thousands of users. It runs on a large cluster and on the cloud.



-  **Thousands** of documented and maintained **tools**
-  **10,000+** users
-  **Free registration** for all users
-  **250 GB** per user by default and more on request
-  **Training** capacities on demand and online



Training

Online training material at training.galaxyproject.org

150+ slides, hands-on tutorials, designed for both self-training and workshops

Technical support with tools, data, and tested workflows on usegalaxy.eu

Topics: Sequence analysis, Transcriptomics, Assembly, Epigenetics, Metagenomics, Proteomics, Variant analysis, Metabolomics, Imaging, Ecology, Computation chemistry, Statistics & machine learning, etc

TlaaS: Training Infrastructure as a Service on usegalaxy.eu

Dedicated on-demand infrastructure for all trainees during workshops

Free, no need for Galaxy maintenance or administration

Official Galaxy Training Materials guaranteed to work

Details on galaxyproject.eu/tiaas



An European project

Distributed compute resources

Pulsar

Openly managed infrastructure via [Github.com/usegalaxy-eu](https://github.com/usegalaxy-eu)

Distributed compute resources



 galaxyproject.eu

 contact@usegalaxy.eu

 gitter.im/usegalaxy-eu/Lobby

 [usegalaxy-eu](https://github.com/usegalaxy-eu)

 [galaxyproject](https://twitter.com/galaxyproject)





Thousands of Tools

Genomics, HTS

Quality control: FastQC, MultiQC, Trim Galore!, etc
Alignment: BLAST, Diamond, etc
Mapping: Bowtie2, STAR, HISAT2, BWA, segemehl, minimap2, etc
Assembly: Unicycler, SPades, Quast, Flye, Canu, etc
Annotation: SortMeRNA, Aragorn, Roary, Prokka, Augustus, KOBAS, Glimmer, antiSMASH, etc
Transcriptomics: FeatureCounts, DESeq2, Trinity, Salmon, etc
RNA: LocARNA, RNAfold, RNAz, RNAplot, etc
Variant Calling: FreeBayes, Gemini, VCFTools, SnpEff, etc
Peak Calling: MACS2, Piranha, PEAKachu, etc
Epigenetics: Bismark, metilene, bwameth, Methyldackel, etc
deepTools, SAM Tools, HicExplorer, Picard, EMBOSS, etc
Metagenomics: MetaPhlAn2, HUMAnN2, VSearch, QIIME, Mothur, MEGAHIT, MetaSpades, etc
Single-cell: Scanpy, Scatter, Seurat, Race-ID, etc
Nanopore: Porechop, Nanopolish, Nanoplot

File and metatools

Get data: USCS, Uniprot
Send data: GenomeSpace Exporter
Convert

Text tools

Text manipulation
 Filter and Sort
 Join, Substract and Group

Proteomics, Metabolomics, Chemistry

OpenMS, PeptideShaker, SearchGUI, MADLIquant, etc
 Jmol Editor, Docking, etc
 OpenBabel, ChemFP, OMG, QED, etc
 ChemicalToolBox

Statistics & Machine learning skit-learn

Climate & Ecology

Copernicus Essential climate variables, etc
 GIS data handling: GDAL suite, etc
 Animal detection on accoustic recording
 Phenology & trend computation

Domain-specific European Galaxies

Domain-specific versions of usegalaxy.eu server
 Set of curated tools, standardized high-quality workflows and training focused on one specific domain
 Dedicated interface but same server on the back with user data

Several European Galaxies to foster communities
 annotation.usegalaxy.eu, cheminformatics.usegalaxy.eu, climate.usegalaxy.eu, clipseq.usegalaxy.eu, ecology.usegalaxy.eu, imaging.usegalaxy.eu, metabolomics.usegalaxy.eu, metagenomics.usegalaxy.eu, nanopore.usegalaxy.eu, plants.usegalaxy.eu, proteomics.usegalaxy.eu, rna.usegalaxy.eu, singlecell.usegalaxy.eu, etc



Virtualization for Sensitive Data

For sensitive (biomedical) data and users with internet limitation

Virtualization of Galaxy via the Galaxy Docker project

Full-fledged Galaxy installation with all dependencies
 Easily extendable bases images with tools, data and workflows

Customized flavors to meet different needs

Installation of community-maintained curated sets of tools, workflows, data and training materials



Data Analysis for any Scientist

No programming knowledge required

Access Galaxy and its tools with a web browser at usegalaxy.eu

Powerful data analyses, to replace Excel habits

Powerful computational infrastructure

Computation automatically performed on a cluster and on the de.NBI cloud

250 GB of storage for analyses, more on request

From beginners to advanced users

Development of novel analyses right in the interface with Interactive Tools via live.usegalaxy.eu: Jupyter lab, RStudio, Ethercalc, cellxgene, etc.

Programmatic access via an API to leverage cluster usage and data storage to scale beyond the visual interface and to automate the analyses



Visualization & Data Interpretation

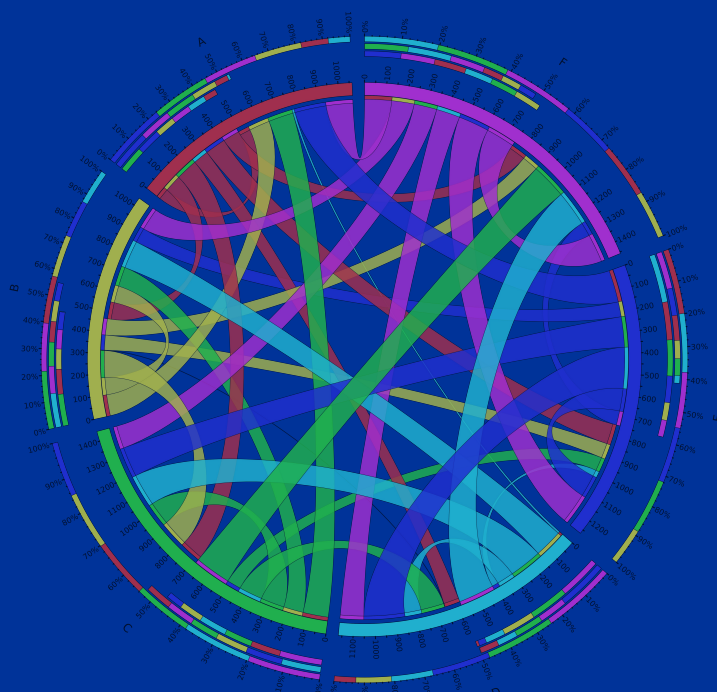
Instantaneous publication-ready visualizations

Charts: bar diagram, line charts, box plots, heatmaps, etc

Scatterplot, Venn diagrams, and more

Interactive visualizations for digging deeper

Integrated Genome Browser, Trackster
JBrowse, Circos, IGV, UCSC, Phinch,
pyGenomeTrack, KRONA, etc



Reproducibility & Transparency

Histories to track analysis

Foundation of reproducibility and transparency in Galaxy, capturing inputs, parameters, and versions of the tools

Sharable with everyone or specific groups

Powerful workflow system

Extraction of workflows from histories or from scratch with drag-and-drop

Versioned, annotable, downloadable and sharable with everyone, no vendor lock-in

Tools and reference data

Fixed versions of thousands of tools, managed by Bioconda and BioContainers

6+TB of reference data providing access to 100+ of reference genomes